Daniel **Yao-Ting Huang**

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Education

University of California, San Diego — *La Jolla, CA*

M.S. in Electrical and Computer Engineering (Intelligent Systems and Robotics/Control) — *Sep. 2024 - Jun. 2026*

- Overall GPA: 3.5/4.0
- Selected Courses: Sensing & Estimation Robotics, Planning & Learning Robotics, Robot Manipulation and Control

National Taiwan University (NTU) — *Taipei, Taiwan*

B.S. in Electrical Engineering — *Sep. 2019 - Aug. 2023*

- Overall GPA: 3.71/4.3(3.64/4.0) Last 60 GPA: 3.82/4.3(3.75/4.0)
- **Teaching Assistant** for Cornerstone EECS Design and Implementation (2023 Spring) - Instructor: Prof. Ho-Lin Chen
- Selected Courses: Robotics, Machine Learning, Deep Learning for Computer Vision.

Publications

In-Hand Manipulation of Articulated Tools with Dexterous Robot Hands with Sim-to-Real Transfer
Soofiyan Atar, **Daniel Huang**, Florian Richter, Michael Yip
arXiv preprint arXiv:2509.23075 (2025). 2025, International Conference on Intelligent Robots and Systems (**IROS'26**)

The N2D Haptic Glove: A Multi-Finger Glove for 2D Directional Force Feedback for Contact Rich Manipulation
Yao-Ting Huang, Jake Honma, Omar Hernandez, Logan Li, Kaitlin Calimbahin, Bryce Hackel, Michael C. Yip
arXiv preprint arXiv:2606.14083 (2026). 2026, under review

Transferring Contact, Not Just Motion: Compliant Grasping Across Dexterous Hands
Soofiyan Atar*, **Yao-Ting Huang***, Michael Yip
arXiv preprint arXiv:2606.15516 (2026). 2026, under review (*Equal contribution)

Research Experience

Advanced Robotics and Controls Lab (ARCLab), UC San Diego — *Advisor: Michael C. Yip*

Graduate Research Assistant — *Oct 2024 – Present*

- **N2D Haptic Glove:** Designed **closed-loop 2-D fingertip force-feedback control** and the full **teleoperation pipeline** for a multi-finger glove driving an Inspire Hand on a Franka arm; in a user study, 2-D feedback achieved the **lowest median force-tracking error** vs. 1-D and no-force-feedback conditions.
- **In-hand manipulation of articulated tools (sim-to-real):** Designed an **augmented tactile fingertip structure** for the Inspire Hand and built the **experimental pipeline** quantifying tool steadiness during dexterous in-hand manipulation.
- **Compliant grasping across dexterous hands:** Built the **joint-torque calibration** and **teleoperated demonstration-collection** pipelines and a **unified hand representation** with force rendering across embodiments (Inspire, Wuji); raised average task success from **0.44 to 0.71** over a DiT-flow baseline, with the largest gain on marker picking.

Robot Learning Lab, NTU — *Shao-Hua Sun*

Undergraduate research student — *July 2022 – Dec 2023*

- Investigated Unsupervised Reinforcement Learning (RL) problems based on SPIRL.
- Designed experiments showing that the learning efficiency will be slower by **1M** steps due to the absence of certain skills.
- Implemented choice-mechanism that allows high-level agent to choose between exploration and exploitation.

Advanced Control Lab, NTU — *Li-Chen Fu*

Undergraduate research student — *Feb 2022 – June 2022*

- Assisted the experiment of the UAV (unmanned aerial vehicle) system in the indoor environment and anchor setup.
- Conducted experiments and developed a **2-D visual odometry** system to achieve indoor UAV odometry in GPS-denied environments using **optical flow** techniques.

Work Experience

Industrial Technology Research Institute (Intern) — *Hsinchu, Taiwan*

Reinforcement Learning, Autonomous system, ROS, Docker — *Aug. 2023 - Dec. 2023*

- Contributed to self-driving automobile group by applying reinforcement learning for behavioral prediction of autonomous vehicles.
- Designed a virtual training environment for reinforcement learning framework based on real-world collected vehicle data.

CAVEDU (Summer Intern) — *Taipei, Taiwan*

Python, Google AIY, Program Education — *Jan. 2016 - Jun. 2017*

- Delivered Python/ML workshops; co-authored teaching kits used by **100+** K-12 students.

Selected Projects

Visual-Inertial SLAM

ECE 276A Sensing & Estimation for Robotics *Jan 2025 - Mar 2025*

- Built stereo pipeline (Shi–Tomasi/optical-flow/RANSAC) with EKF fusion; update time **<50 ms** on recorded sequences.
- Tuned noise models and outlier rejection; achieved drift decreasing ∼**80%** vs IMU-only across **3** evaluation runs.
- Implemented robust keyframe logic to sustain tracking under fast motion.

Motion Planning Algorithms

ECE 276B Planning & Learning for Robotics *Apr 2025 - May 2025*

- Developed **Weighted A*** and KD-accelerated **Connect-RRT** planners for 3-D obstacle maps; achieved **26×** NN-query speed-up with cKDTree.
- Evaluated seven maps: Weighted A* (w = 2) ran 23 % faster than optimal A* with only 3 % longer paths.
- Produced comparative study to analyze runtime, memory, and path quality across search-based and sampling-based methods.

Magic Hand — Real-Time RGB-D Arm Pose Guided Laser Pointer.

2023 MakeNTU [Demo] *Apr 2023 - May 2023*

- Applied **YOLOv5** on RGB-D frames from **Intel RealSense** camera to localize arm joints and estimate depth for laser-pointer steering in real time.
- Optimized inference with **optical-flow** tracking to reduce per-frame compute and sustain **60+ FPS** on a laptop CPU.
- Delivered an **award-winning demo** with robust, low-latency pointing and smooth user interaction.

Sushiro-Bot

CSIE5047 Robotics. [Demo] *Oct 2022 - Dec 2022*

- Developed a 7-axis robot arm for making Nigiri-sushi in **80s** and Tekka-maki in **5mins**, incorporating computer vision techniques for precision.
- Used computer vision techniques to automatically evaluate the quality of the sushi and calculate the best gripping point for soft ingredients
- Designed an algorithm that calculated the distribution of the rice and gave a suggested spot to fill the rice.

Awards & Competitions

2023	**3rd prize of Enterprise prize & Best Application Award**, MakeNTU	*Taipei, Taiwan*
2021	**3rd prize of Interactive Technology**, NIICC	*Taipei, Taiwan*
2021	**4th prize of Enterprise prize**, MakeNTU	*Taipei, Taiwan*
2020	**Special prize of Enterprise prize**, Meichu Hackathon	*hsinchu, Taiwan*
2020	**Special prize of Enterprise award**, MakeNTU	*Taipei, Taiwan*

Leadership & Volunteer experience

2022 MakeNTU

Event General Coordinator *Aug 2021 - May 2022*

- Orchestrated a national competition with over **34 teams** and **134 undergraduates**, leading marketing and coordination efforts.
- Enhanced event visibility, contributing to a **20%** increase in participant engagement.

Academic Department of NTUEE Student Association

Manager & Course lecturer *Sep 2020 - Aug 2023*

- Facilitated the usage and maintenance of machine tools for student projects.
- Managed inventory, ensuring the availability of modules and microcontrollers for the whole department (over **200** students).
- Delivered lectures on the fundamental use and advanced techniques of **AutoCAD** and **Fusion360**.

Computer program consulting service

Volunteer *Sep 2020 - Jan 2021*

- Provided weekly consultation services to assist students from various departments in overcoming challenges with programming languages such as C++ and Python.

Skills

Robot Learning	PyTorch, LeRobot, ACT, Diffusion Policy, imitation learning, demonstration dataset design, Isaac Lab, MuJoCo
Dexterous Manipulation	Inspire Hand, Wuji Hand, sim-to-real, teleoperation and hand retargeting, tactile sensing and calibration, force control
Embedded & Mechanical	Teensy, Jetson, sensor characterization, 3D CAD (Fusion, Onshape), compliant mechanism design
Software & Systems	Python, C/C++, ROS 2, Linux, Docker, multi-sensor time synchronization

Academic Services

Reviewer: IEEE ICRA 2027, IEEE/RSJ IROS 2026